

ON THE RIGIDITY OF THE MINIMUM-MODULUS PATTERN: A CIRCLE-ROTATION READING

CLAUDE

ABSTRACT. The minimum moduli $\mu_n^{(c)}$ of a family of characteristic polynomials attached to $\tau(p_n)$ oscillate, and the locations of their extrema obey a strikingly rigid arithmetic pattern. This note argues that the rigidity is not special to τ : granted a single dominant complex frequency, it is the universal rigidity of the return times of an irrational circle rotation (the three-gap theorem), and the specific pattern is a two-term continued fraction made visible. The genuinely open questions are isolated at the end. The note is self-contained; the two analytic inputs it borrows are stated as facts in §2 and proved in a companion note.

1. SETUP AND NOTATION

Let τ be the Ramanujan tau function, defined by $\Delta(q) = q \prod_{k \geq 1} (1 - q^k)^{24} = \sum_{m \geq 1} \tau(m) q^m$, and let p_n denote the n -th prime. Fix the sequence $h_0 = 1$, $h_n = \tau(p_n)$ for $n \geq 1$.

The recurrence and its data. Equation (D) of the manuscript,

$$n h_n = \sum_{r=1}^n j_r h_{n-r} \quad (n \geq 1),$$

is a bijection between sequences $(h_n)_{n \geq 0}$ with $h_0 = 1$ and *data sequences* $(j_r)_{r \geq 1}$. In generating-function form, with $H(t) = \sum_{n \geq 0} h_n t^n$ and $J(t) = \sum_{r \geq 1} j_r t^r$, it reads $tH' = HJ$, i.e. $J = t(\log H)'$.

The deformation. For a parameter c , prepend c to the data sequence, $j^{(c)} = (c, j_1, j_2, \dots)$, and run (D) forward from $h_0 = 1$ to obtain the deformed sequence $h^{(c)} = (h_n^{(c)})_{n \geq 0}$. Equivalently the generating function is multiplied by an exponential,

$$H^{(c)}(t) = e^{ct} G(t), \quad G(t) = \exp \int_0^t J(s) ds,$$

so that $h_n^{(c)} = [t^n] e^{ct} G(t)$. Throughout we take $c = 1$, the case in the computational run.

The spectra. To each n the manuscript attaches the monic degree- n polynomial (Theorem 2.3)

$$\chi_n^{(c)}(x) = \sum_{r=0}^n \binom{n}{r} r! h_r^{(c)} (-1)^r x^{n-r},$$

and writes $\mu_n^{(c)} = \min\{|\zeta| : \chi_n^{(c)}(\zeta) = 0\}$ for the modulus of its root nearest the origin. It is the sequence $(\mu_n^{(c)})_n$, plotted against n , whose oscillation is at issue.

2. THE TWO ANALYTIC INPUTS

The argument below borrows exactly two facts. Both are established in the companion asymptotics note; here they are simply named, with the constants defined as they arise.

Input 1 (dominant singularity). Since $J = tH'/H$, the singularities of G are the zeros of $H(t) = 1 + \sum_{n \geq 1} \tau(p_n)t^n$. The zero nearest the origin is a complex-conjugate pair $t_0, \bar{t}_0$; writing it in polar form

$$t_0 = \rho e^{i\theta}, \quad \rho = |t_0| \approx 0.0435, \quad \theta = \arg t_0 \approx 0.2432 \pi,$$

this defines the modulus ρ and the *argument* θ . (Numerically t_0 is stable under truncation of H , and ρ is corroborated by the coefficient growth $[[t^n]G]^{1/n} \rightarrow 1/\rho$.)

Input 2 (coefficient form and the minima). By singularity analysis [12, 13], a dominant conjugate pair at $t_0 = \rho e^{i\theta}$ forces the leading-order shape

$$h_n^{(c)} = C \rho^{-n} \cos(n\theta + \varphi) + O(\rho_2^{-n} \text{poly}(n)), \quad \rho_2 > \rho,$$

where $\rho_2 \approx 0.078$ is the modulus of the next-nearest zero of H , so the error term is smaller than the main term by a factor $(\rho/\rho_2)^n \approx 0.56^n \rightarrow 0$. Here $C > 0$ is a real constant and φ is a fixed *phase offset* (it absorbs the argument of the complex amplitude attached to the singularity and the shift contributed by the e^{ct} factor; only its value, not its origin, matters below); a slowly varying algebraic factor n^β may multiply the main term and has been absorbed into it schematically. The relation says that the cosine term is the dominant asymptotics, with everything else of strictly smaller exponential order — it is *not* a claim that the ratio of the two sides tends to 1, which would fail at the zeros of the cosine. Because $\chi_n^{(c)}$ is monic, the product of its roots is $n! h_n^{(c)}$, and with the bulk of the spectrum varying smoothly in n the variation of $\mu_n^{(c)}$ tracks $|h_n^{(c)}|$. Consequently $\mu_n^{(c)}$ dips toward zero exactly at the sign changes of $h_n^{(c)}$, i.e. where the phase

$$n\theta + \varphi \text{ is nearest an odd multiple of } \frac{\pi}{2}.$$

3. THE EMPIRICAL REGULARITY TO BE EXPLAINED

A run to $n = 500$, followed by a difference-table analysis of the lower and upper envelopes of $(\mu_n^{(1)})_n$, reports the following. The successive local minima occur at indices whose consecutive gaps are drawn from exactly *two* values, $\{4, 5\}$; the gap 5 occurs on a strict period-9 schedule; and the second and third differences of the gap sequence are perfectly periodic with period 9 (no jitter anywhere). One full block is

$$8 \times 4 + 1 \times 5 = 37,$$

containing 9 minima, so the pattern recurs with period 37 in n ; the upper envelope is the same, phase-shifted. The average gap is $37/9 = 4.111\dots$, and an independent Fourier analysis of the same data places its dominant peak at period 4.10. So a discrete method (difference tables) and a spectral method (FFT) agree on a fast period ≈ 4.1 and a long period 37.

4. THE STRICTNESS IS THE FINGERPRINT OF A CIRCLE ROTATION

A sinusoid sampled at the integers would not produce gaps from exactly two values with the 5 on a strict period-9 schedule and perfectly periodic higher differences; it would jitter. The pattern of §3 — two gap lengths, one inserted quasi-periodically — is the signature of a *Sturmian* (mechanical) word [1, 2], and Sturmian words are exactly the symbolic codings of an irrational rotation of the circle.

Input 2 makes the rotation explicit. The minima of $\mu_n^{(c)}$ fall where the phase $n\theta + \varphi$ is nearest an odd multiple of $\pi/2$; tracking them is tracking the returns of the orbit $\{n(\theta/\pi) + \beta\}$ near $\frac{1}{2}$ on the circle, an equidistributed orbit in the sense of Weyl [9, 10]. The *three-gap theorem* — conjectured by Steinhaus, first proved by Sós, Surányi, and Świerczkowski [3, 4, 5, 6] — states that the gaps between successive returns of any irrational rotation take at most three distinct values, arranged quasi-periodically. Here they take two, $\{4, 5\}$. So the rigidity is not a special fact about τ : it is the universal rigidity of rotation return times. Once one grants the single dominant frequency θ of Input 1, the strict two-value pattern is *forced*; nothing else was available.

5. THE SPECIFIC PATTERN IS THE CONTINUED FRACTION MADE VISIBLE

The average gap is $37/9 = 4.111\dots$, and as a continued fraction

$$\frac{37}{9} = [4; 9] = 4 + \frac{1}{9}.$$

That two-term expression *is* the observed law: the integer part 4 is the base gap; the partial quotient 9 is the period of the correction (one extra step — a 5 — every nine). The minima locations are, to this order, a Beatty sequence $[k(37/9) + \beta]$ [7, 8]. The fast period π/θ predicted by Input 2 equals $1/0.2432 \approx 4.11$, matching $37/9$. So the puzzle of why an $8 \times 4 + 5$

block repeats with period 37 collapses to a two-term continued fraction; the Fourier peak at 4.10 is the blurred analytic shadow of this crisp arithmetic object, which is why the difference-table view looks far more rigid than a spectral peak has any right to.

6. WHERE THE INTERESTING NUMBER THEORY ACTUALLY LIVES

Two facts remain unexplained by the model.

First, that $H(t) = 1 + \sum_n \tau(p_n)t^n$ has a single, clean, well-isolated dominant complex zero at all (Input 1). That one conjugate pair should govern everything is a genuine statement about the analytic structure of a *prime-indexed* τ series — an object with no modularity, no functional equation, and no Euler product. The dominant-singularity principle [12, 13] says what such a zero *would* control, not why this series should have so clean a one. I know no reason.

Second, and more pointed: the value of the rotation number θ/π and its proximity to $9/37$. Is it *exactly* $9/37$ or merely very close? This is decidable, and the meaning hinges on it.

- If θ/π is *irrational* with continued fraction $[0; 4, 9, a_3, \dots]$ and a_3 large, then $9/37$ is an excellent rational approximant, the coding is Sturmian [1, 2], and near $n \sim 37a_3$ a *second-order defect* must appear — a block with seven or nine 4's rather than eight, or two consecutive 5's. The run to $n = 500$ shows about thirteen clean cycles with no such defect, which already forces a_3 to be large: $|\theta/\pi - 9/37| \lesssim 1/(37 \cdot 500) \approx 5 \times 10^{-5}$.
- If θ/π were *exactly* $9/37$, the pattern would be exactly period-37 forever. That would be astonishing: it would place the nearest zero of a prime-indexed τ series at a rational multiple of π , demanding an arithmetic explanation.

My honest bet is the first — irrational, $[0; 4, 9, a_3, \dots]$ with a_3 large — because there is no reason a transcendental-looking θ assembled from τ at primes should be a rational multiple of π , and every real number masquerades as a small-denominator rational when known to only four digits. I hold it loosely; the second case, if true, would be worth a paper.

7. HOW TO SETTLE IT

Pin θ down to eight or ten digits (rather than the four currently in hand), form θ/π , and read its continued fraction [8]. If it begins $[0; 4, 9, a_3, \dots]$, then a_3 *predicts the exact n at which the first second-order defect must occur* — a prediction tested by extending the run past $n = 500$ to that point. A defect at the predicted place proves the rotation-number account and shows θ/π irrational; no defect to all computable precision would be the surprising case. In the language of circle dynamics this is just reading the rotation number and its convergent denominators off the orbit [11].

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